

Mixed identities and Neumann's Lemma

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joint work with Michael Pinsker

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Outline

① Background

Equations in groups

Oligomorphicity

Mixed identities

② On algebraic closure

③ The result

④ Strategy

⑤ Bibliografy

Words with constants

G := a group.

Definition (Word with constants)

A **word with constants** for G in r many variables is

$$w(x_1, \dots, x_r) := \gamma_n x_{\iota_n}^{\epsilon_n} \gamma_{n-1} x_{\iota_{n-1}}^{\epsilon_{n-1}} \dots \gamma_1 x_{\iota_1}^{\epsilon_1} \gamma_0$$

for $\gamma_0, \dots, \gamma_n \in G$, $\epsilon_1, \dots, \epsilon_n \in \{-1, 1\}$, and $\iota_1, \dots, \iota_n \in \{1, \dots, r\}$.

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I.e., a **term** in $\mathcal{L} := \{\cdot, {}^{-1}, (\gamma)_{\gamma \in G}\}$.

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We ask that w is **reduced**: $\iota_{j-1} = \iota_j$ and $\epsilon_{j-1} = -\epsilon_j \Rightarrow \gamma_j \notin Z(G)$.

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w is **singular** if “forgetting constants” and reducing we get 1:

$$y^{-1} x \gamma_2 x^{-1} \gamma_1 y \xrightarrow[\text{constants}]{\text{forget}} y^{-1} x x^{-1} y \mapsto y^{-1} y \mapsto 1.$$

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Otherwise, w is **regular** (a.k.a. non-singular).

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w defines a **word map**: $w : G^r \rightarrow G \ (g_1, \dots, g_r) \mapsto w(g_1, \dots, g_r)$.

- when can we **solve** w ?

$$\exists g_1, \dots, g_r \in G \text{ s.t. } w(g_1, \dots, g_r) = 1 ;$$

- when is w a **mixed identity**?

$$\forall g_1, \dots, g_r \in G \ w(g_1, \dots, g_r) = 1 .$$

👁 These notions are in tension!

$$\forall \bar{g} \in G \ w(\bar{g}) = 1 \Rightarrow \text{for } \gamma \in G \setminus \{1\}, \nexists \bar{g} \in G \ \gamma w(\bar{g}) = 1.$$

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Solving equations

Conjecture (Klyachko and Thom 2017)

G a group, $w(x_1, \dots, x_r)$ **regular**.

Then, there is $H \supseteq G$ s.t. $\exists h_1, \dots, h_r \in G$ $w(h_1, \dots, h_r) = 1$.

- $r = 1$ this is the Kervaire-Laudenbach Conjecture:
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Interest in finding groups for which $w(G^r)$ is large (possibly $= G$).

(because $\gamma \notin w(G^r) \Leftrightarrow 1 \notin \gamma^{-1}w(G^r)$)

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Theorem (Adeleke and Holland 1994)

$G := \text{Aut}(\mathbb{Q}; <)$. w *with no constants and non-trivial*.

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Theorem (Mycielski 1987; Lyndon 1990)

$G := \text{Sym}(\mathbb{N})$. w **with no constants** and $w \neq x^n$ ($n \geq 2$).

Then, $w(G^r) = G$.

An interlude on oligomorphicity

$\text{Aut}(\mathbb{Q}; <)$ and $\text{Sym}(\mathbb{N})$ are both oligomorphic:

Definition (Oligomorphic)

$|\Omega| = \aleph_0$. $G \curvearrowright \Omega$ is **oligomorphic** if for all $n \in \mathbb{N}$, $G \curvearrowright \Omega^n$ has finitely many orbits in the action $g \cdot (a_1, \dots, a_n) \mapsto (ga_1, \dots, ga_n)$.

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Many examples are **homogeneous**: every isomorphism between finite substructures of M extends to an automorphism.

Fraïssé:

$$\left\{ \begin{array}{c} \text{classes of finite substructures (ages) of} \\ \text{homogeneous structures} \end{array} \right\} \leftrightarrow \left\{ \begin{array}{c} \text{Fraïssé} \\ \text{classes} \end{array} \right\}.$$

Fraïssé:* hereditary class $\mathcal{C}$ with the **amalgamation property**:

(AP) for $A, B_0, B_1 \in \mathcal{C}$, embeddings $f_i : A \rightarrow B_i$ ($i \in \{0, 1\}$),
there is $C \in \mathcal{C}$ and embeddings $g_i : B_i \rightarrow C$ ($i \in \{0, 1\}$) s.t.
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Often, we specify homogeneous structures by their age:
random graph, generic triangle-free graph, random poset. . .

*Assuming language is purely relational.

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Definition (MIF and lawless)

$w(x_1, \dots, x_r)$ is a **mixed identity** if $w(G^r) = 1$.

👁 Mixed identities describe the structure of a group/elements:

$$\forall x \ x^2 = 1 \quad \text{or} \quad \forall x \ xcx^{-1}c^{-1} = 1$$

💡 we would expect “rich group” $\Rightarrow$ “few” mixed identities.

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Theorem (Melles and Shelah 1994)

M a saturated structure[†] $\Rightarrow \text{Aut}(M)$ is lawless.

[†]**saturated**: types over subsets $A \subseteq M$ with $|A| < |M|$ are realised in M .

Mixed identities in the oligomorphic/homogeneous context

Several automorphism groups of homogeneous structures are MIF:

- random graph, generic triangle free graph, Urysohn space

(Etadialiabadi, Gao, Le Maître, and Melleray 2021);

- transitive with free amalgamation

(Ghadernzhad and de la Nuez González 2019; Bodirsky, Schneider, and Thom 2024);

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- $\text{Sym}(\mathbb{N})$: $[x, t]^6$ where t is a transposition
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Sometimes we have mixed identities!

- $\text{Sym}(\mathbb{N})$: $[x, t]^6$ where t is a transposition
(Hull and Osin 2016);
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Mixed identities in the oligomorphic/homogeneous context

Several automorphism groups of homogeneous structures are MIF:

- random graph, generic triangle free graph, Urysohn space
(Etadadialiabadi, Gao, Le Maître, and Melleray 2021);
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In the first two cases, we know all mixed identities are singular!

- $\text{Sym}(\mathbb{N})$: all mixed identities are singular
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Conjecture (Bodirsky, Schneider, and Thom 2024)

$G \curvearrowright \Omega$ oligomorphic $\Rightarrow$ all mixed identities of G are singular.

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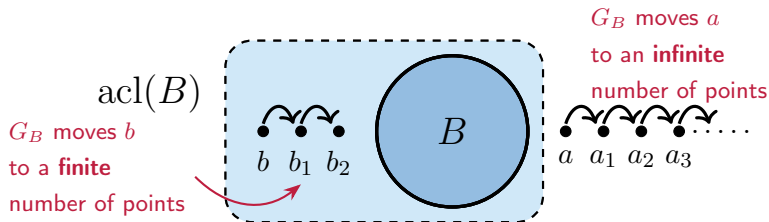
We want general conditions that imply all mixed identities are singular.

Algebraic closure

Definition (algebraic closure)

$G \curvearrowright \Omega$. $B \subseteq \Omega$ finite.

$$\text{acl}(B) := \{a \in \Omega \mid a \text{ has finite } G_B\text{-orbit}\}.$$



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Lemma (II. M. Neumann 1976)

$G \curvearrowright \Omega$. $A, B \subseteq \Omega$ finite and $A \cap \text{acl}(\emptyset) = \emptyset$.

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Theorem (Abért 2005)

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Our result

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- automorphism groups of homogeneous structures whose age has the **strong amalgamation property**:
 - recovers most previous results;
 - confirms the Bodirsky, Schneider, and Thom 2024 conjecture for several cases beyond previous techniques:
 ($\mathbb{Q}; <$) and reducts, Droste's 1985 2-homogeneous trees of negative type, homogeneous local order $S(2)$, homogeneous B, C, D relations;

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- Weakly branch groups (include Grigorchuk's group).
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- Homeomorphism groups of: Cantor set 2^ω , Hilbert cube $[0, 1]^\omega$, any manifold of dimension ≥ 1 ;
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Proof strategy[‡]

For $g \in G$, $a_0 \in \Omega$, the images of a_0 with respect to subwords of w create a sequence $(a_0, a_1, a'_1, \dots, a_n, a'_n, a_{n+1})$ with

$$g \cdot (a_1, \dots, a_n) = (a'_1, \dots, a'_n) .$$

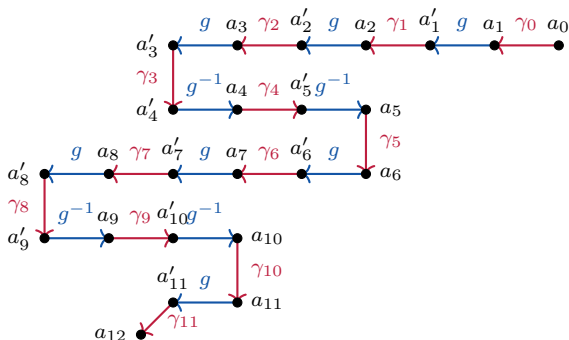
[‡] w with one variable.

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$$w(x) = \gamma_{11}x\gamma_{10}x^{-1}\gamma_9x^{-1}\gamma_8x\gamma_7x\gamma_6x\gamma_5x^{-1}\gamma_4x^{-1}\gamma_3x\gamma_2x\gamma_1x\gamma_0$$

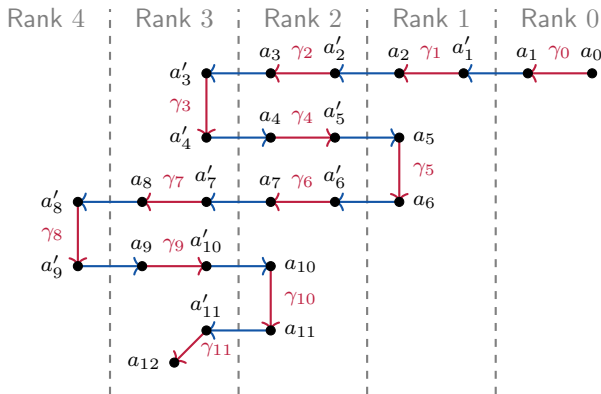


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Rank elements according to their position in $\mathbb{Z}$.

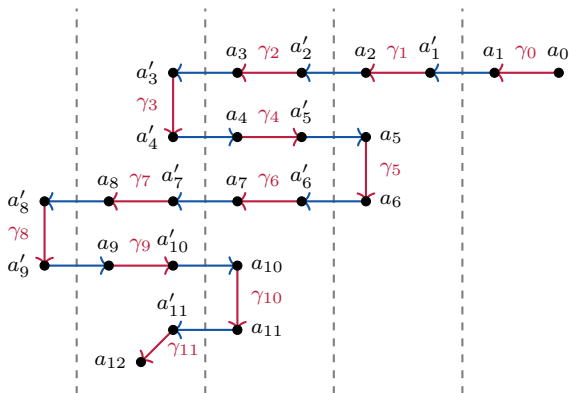


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We want g so that elements of different ranks are independent!

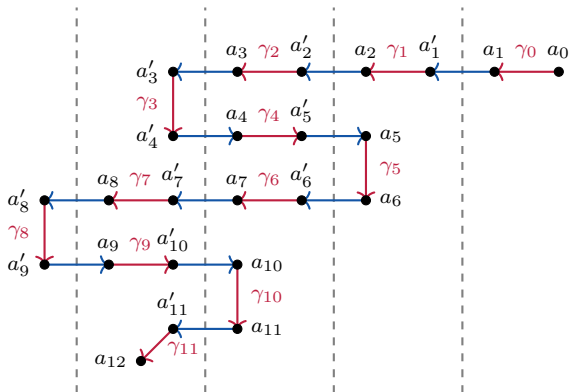


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We cannot guarantee elements of same rank to be distinct!

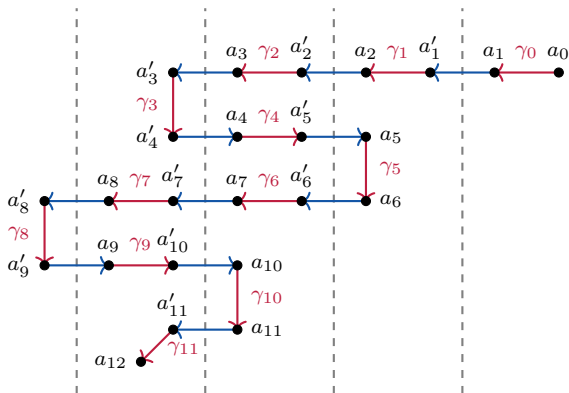


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Use **Neumann's Lemma** to guarantee independence (looking ahead!)

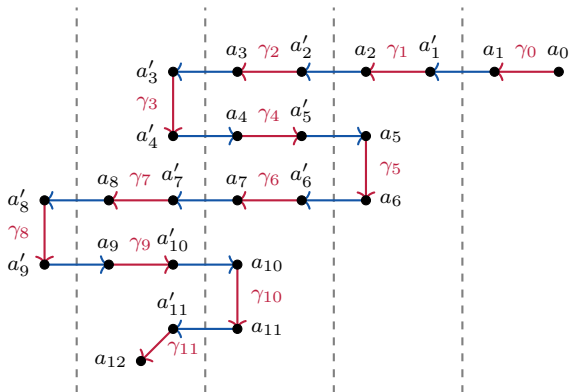


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Induction proof: challenge is balancing inductive hypothesis!



The real Theorem

With algebraicity, we need a further geometric properties and a strengthening of Neumann's Lemma.

Definition

Let $G \curvearrowright \Omega$. (Ω, acl) is **higher Neumann** if for each $B, C \subseteq \Omega$ finite, $a \in \Omega$ such that $\text{acl}(a) \cap \text{acl}(B) = \text{acl}(\emptyset)$, and $\Sigma \subseteq G$ finite, $g \in G_B$ such that

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acl forms a **pregeometry** if it satisfies:

(EXCHANGE) if $a \in \text{acl}(A \cup \{b\}) \setminus \text{acl}(A)$, then $b \in \text{acl}(A \cup \{a\})$.

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It is **modular** if for all closed $A, B \subseteq \Omega$,

$$\dim(A \cup B) = \dim(A) + \dim(B) - \dim(A \cap B) .$$

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Theorem (MP 2026)

Let $G \curvearrowright \Omega$ be s.t. acl forms an infinite-dimensional higher Neumann modular pregeometry. Then all mixed identities of G are singular.

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Let $G \curvearrowright \Omega$ be s.t. acl forms an infinite-dimensional higher Neumann modular pregeometry. Then all mixed identities of G are singular.

Applies to $\text{GL}(\aleph_0, \mathbb{F}_q)$ and $\text{PGL}(\aleph_0, \mathbb{F}_q)$, recovering results of Bradford, Schneider, and Thom 2023.

Thank you!

Recap:

- Mixed identities are a way to describe structure in groups;
- We prove: all mixed identities are singular for groups with no algebraicity;
- Theorem covers many classes of groups for which people are studying their mixed identities.

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